power as input to drive. The electric current required for this vehicle is obtained from sunlight through light voltage cells. These batteries convert solar power into electricity.

It has been incorporated into alternative energy and vehicle technologies. Most solar vehicles are used for car races. Nowadays, solar car systems are designed for the use on roads.



Fig. 5.15 Solar vehicle

When sunlight falls on photo voltaic cell, it induces electrons and converts it as electrical energy. Photo voltaic cells consist of silicon and alloys such as indium, gallium and nitrogen, with huge quantity of silicon.



Fig. 5.16 Solar vehicle

5.5.4 Battery

Storage battery packs in a solar system carries 250 miles (400 km) without the rays of sun. More cars have enough power to allow them to travel at a speed upto 60 miles (97 km/h).

5.6 Electric traction

The electric traction is divided into two types.

1. The Urban train and
2. The Sub-urban train.

SUIN2Z

1. Urban train

In this way the drives are given necessary power for two ways, as shown in figure 5.17.

1. Getting power from the overhead lines.
2. Getting power from the diesel drive.

The collector is mounted and the current passes through the overhead transmission line.



Fig. 5.17 Electric train

2. Sub-urban train

Sub-urban trains are used to travel short distances. The train is called the local train. The sub-urban train can be used to stop and park to a distance of 2 to 5 km intervals.

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